Burgernomics After 25 Years: The Gap Closes Through Burger Prices

The Big Mac index, 2000-2026, and price levels since 1880

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Abstract

Over 25 years of the Economist's Big Mac index, a currency that looks overvalued in burger terms has been no more likely to depreciate against the dollar than any other. For floating currencies the slope of future exchange-rate changes on the published valuation is zero at horizons of one to five years, and out of sample the index forecasts exchange rates worse than a random walk, even after each currency's persistent valuation is removed. The burger gaps that do close, about a fifth of a raw gap within five years, close through local burger prices, as menu prices catch up with past currency moves. National price levels under floating rates adjust through the exchange rate instead, which in 17 countries over 1974-2020 accounts for −0.92 of a five-year real exchange rate slope of −1.08. Under the classical gold standard relative price levels did the closing, so the Big Mac gap adjusts the way national price levels did when exchange rates were fixed.

1 The index and the test

The Big Mac index compares the dollar price of a Big Mac across countries. If a burger costs more in dollars in country i than in the United States, the currency is called overvalued, and the implied forecast is that it will fall. We use the Economist's published data, which cover 55 economies besides the United States at 45 survey dates from April 2000 to July 2026. Let

$$v_{it} = \ln P_{it} - \ln e_{it} - \ln P_t^{\$},$$

where P_{it} is the local burger price, e_{it} is local currency per dollar and $P_t^{\$}$ is the US price, so that $v > 0$ means overvalued. Over a horizon of h years the valuation changes by $\Delta v = -\Delta \ln e + (\Delta \ln P - \Delta \ln P^{\$})$, the sum of an exchange-rate part and a relative burger-price part. We regress the change in v and each part on v at the start. A slope of -1 means the gap closes in full; a negative slope on the exchange-rate part means overvalued currencies depreciate.

We drop eleven currencies whose annual change against the dollar has a standard deviation below 2 percent, which are pegged in practice: those of the Gulf states, Azerbaijan, Guatemala, Hong Kong, Jordan and Nicaragua. In the main sample we also drop seven currencies whose annual change against the dollar averaged more than 12 percent in absolute value (Argentina, Brazil, Egypt, Lebanon, Turkey, Ukraine and Venezuela), most of them economies with episodes of high inflation; results with them included are reported in the text. Observations are matched at the survey date closest to $t + h$ within 75 days, and standard errors are clustered by survey date.

2 Does the index predict exchange rates?

	Raw valuation			Valuation relative to country mean		
Horizon	Gap	Exchange rate	Burger price	Gap	Exchange rate	Burger price
1 year	−0.05 (0.02)	0.00 (0.02)	−0.05 (0.01)	−0.24 (0.05)	−0.10 (0.05)	−0.14 (0.02)
2 years	−0.10 (0.02)	0.00 (0.02)	−0.10 (0.01)	−0.41 (0.05)	−0.18 (0.05)	−0.23 (0.02)
5 years	−0.22 (0.02)	0.00 (0.02)	−0.22 (0.01)	−0.76 (0.06)	−0.36 (0.07)	−0.40 (0.04)

Table 1: Slopes of the change in the Big Mac valuation over the horizon, and of its exchange-rate and relative burger-price parts, on the valuation at the start. 37 floating currencies without the seven high-volatility ones; 1,346, 1,281 and 1,090 observations at one, two and five years. The right-hand columns add country fixed effects. Standard errors clustered by survey date in parentheses.

	1 year	2 years	5 years
Raw valuation, 44 floating currencies	−0.00	−0.05	−0.22
Valuation relative to country mean to date, 44 floating currencies	−0.02	−0.17	−0.36
Valuation relative to country mean to date, without high-volatility	−0.10	−0.23	−0.80

Table 2: Out-of-sample R^2 of exchange-rate forecasts relative to no change, $1 - \sum(\text{error})^2 / \sum(\text{change})^2$, for forecasts made from 2010 onward with regressions estimated on data available at each forecast date. Negative values mean the forecast is worse than no change. Each regression uses only observations whose horizon had ended by the forecast date, and the country mean is the average valuation over all surveys up to the forecast date, required to cover at least six surveys.

Published valuations carry no information about future exchange rates. In Table 1, the slope of the exchange-rate part on the published valuation is zero to two decimals at every horizon, so a currency that looks 20 percent overvalued is expected to move against the dollar by the same amount as one that looks fairly valued. Gaps close slowly, by about a fifth in five years, and entirely through burger prices. Including the seven high-volatility currencies does not change this; the five-year slope of the exchange-rate part is then 0.13 with a standard error of 0.09, the wrong sign for the index.

Much of the raw valuation is permanent. Poorer countries have cheaper burgers year after year, for the Balassa-Samuelson reasons that the Economist's GDP-adjusted index addresses. Measured against each country's own average, the gap closes faster, three quarters in five years, and the exchange rate does about half of the closing. That version uses each country's full-sample mean, which a forecaster would not have had, and Table 2 shows that the information does not survive without it. Each forecast uses only data available at the forecast date, including the country means used to remove the persistent part of the valuation. Exchange-rate forecasts are worse than a no-change forecast at every horizon, and the shortfall grows with the horizon.

Earlier evaluations were more favorable to the index. Clements, Lan and Seah (2012) assess its first two decades, and an earlier working-paper version of their study reports that after a bias correction it forecasts at least as well as a random walk except at short horizons, and that exchange rates respond more than burger prices to mispricing. Fifteen more years of data and many more emerging-market currencies leave the index with no forecasting value and reverse the ordering of the two channels.

Why burger prices move. Burger prices could move because of noise in a single surveyed price, such as a promotion that ends, or McDonald's managing a price it knows is watched. To separate these from economic adjustment, we split each country's change in valuation over the past year into its exchange-rate and burger-price parts and regress the next year's relative burger-price change on both, controlling for the valuation a year earlier. For the 37 floating currencies, next year's burger price undoes 0.17 of last year's exchange-rate move (a coefficient of −0.17 with a standard error of 0.04, from 1,095 observations matched to the survey in the same calendar month a year earlier), so after a depreciation the local menu price catches up with a lag. Last year's burger-price move is not undone, with a coefficient of −0.03 (0.05) and a year-to-year correlation of annual burger-price changes of 0.04, which is hard to reconcile with temporary promotions or survey noise. In the seven high-volatility currencies the catch-up is 0.66 (0.20) of the past exchange-rate move. Most of the burger-price channel is therefore lagged pass-through of currency moves into prices set in local currency. Deliberate management of the published price would tend to keep gaps open, and the best-known reported case, government pressure on McDonald's pricing in Argentina around 2011-12, falls in the excluded high-volatility group. In the pegged currencies, where the exchange rate barely moves, the five-year slope with country fixed effects is −1.14, all of it through burger prices, and there past burger-price moves are partly reversed (−0.22, standard error 0.15), so noise may matter more for them.

3 Whole price levels behave differently

For national consumer price levels, the Macrohistory Database of Jordà, Schularick and Taylor (2017) gives the same decomposition since 1880 (Table 3). Under floating rates since 1974 the gap closes through the exchange rate, with the nominal rate accounting for −0.92 of a five-year slope of −1.08, as Eichenbaum, Johannsen and Rebelo (2021) find for inflation-targeting countries. Under the classical gold standard, with nominal rates fixed, relative price levels did all the closing, somewhat more slowly, with a least-squares half-life of 4.4 years against 2.7 under floating. Bretton Woods sits in between, with adjustment through the occasional devaluations and slower reversion. These least-squares half-lives are short of their bias-corrected

Regime	Real rate	Nominal rate	Relative prices	Half-life (years)
Gold standard, 1880-1913	−0.66 (0.07)	−0.01 (0.05)	−0.65 (0.06)	4.4
Interwar, 1925-1938	−1.04 (0.12)	−0.93 (0.19)	−0.11 (0.09)	2.1
Bretton Woods, 1950-1971	−0.54 (0.05)	−0.39 (0.06)	−0.14 (0.04)	4.1
Floating, 1974-2020	−1.08 (0.08)	−0.92 (0.10)	−0.16 (0.04)	2.7

Table 3: Five-year changes in the consumer-price real exchange rate against the dollar, and its nominal and relative-price parts, regressed on its level at the start, with country fixed effects and standard errors clustered by year; start and end years lie inside the regime. Seventeen countries from the Jordà-Schularick-Taylor Macrohistory Database (sixteen under the gold standard, for which Ireland has no data). The half-life is the least-squares estimate from a pooled annual AR(1) within the regime.

values. Over the floating period the median-unbiased half-life is 3.7 years with a 90 percent interval of 1.9 to 10.1 years, computed in the manner of Andrews (1993) by simulating the pooled within estimator with shocks that have the cross-country correlation of the estimated residuals, on average 0.65. Treating the countries as independent would give a narrower interval.

Even where exchange rates float, a Big Mac behaves like a price level under the gold standard. A Big Mac combines traded ingredients with local labor and rent, and Parsley and Wei (2007) use its known recipe to separate the two. Much of the burger gap is therefore a gap in local costs, which the exchange rate need not close.

4 Scope

The Big Mac sample is short, 26 years, and the regressions with country fixed effects build some reversion toward the sample mean into the estimates, as do the regime regressions, most of all in the short interwar window. Each observation is a single good surveyed twice a year, and the pass-through test cannot rule out noise or managed pricing in individual episodes; a comparison of Big Mac inflation with each country's consumer price inflation would sharpen it. The historical series use the Macrohistory Database's exchange rates and consumer price indices without adjustment, and the regime dates are conventional.

References

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